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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Hokkien Mee Is Purple
Selected Domain	Domain 1: Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 4: Quality Education (main; target 4.4). Supporting: SDG 8 (targets 8.2, 8.5).
Project / Solution Name	Varity
Name (Leader)	Tay Ee Zhan
Email & Phone (Leader)	eezhan0805@gmail.com / +60 11-6561 6811
University Affiliation(s)	Universiti Malaya (both members)
Number of Team Members	2 (Tay Ee Zhan, Goh Sheng Kuan)

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Aina is a junior Python developer in Malaysia (a synthetic example). She hears that AI is changing tech jobs. She cannot tell which skills to deepen, which to drop, and what to learn next.

In Malaysia, 56% of professionals already use AI tools at work, yet only 26% have received formal training (Hays Asia Salary Guide 2025, cited in the Domain 1 brief). MOSTI has said that more than 600,000 workers will need reskilling within three to five years (Domain 1 brief).
Our user is a junior or mid-level tech developer, or a career switcher, in Malaysia. This user meets three problems:
1. Skill lists carry no date, no source and no measure of doubt.
2. Hype hides the signal. In Hacker News hiring posts, the share that mention AI rose from 6.4% in January 2019 to 54.6% in September 2026. Demand for the LLM / GenAI skill itself showed no clear change: 566 mentions in 3,714 posts in the last 12 months against 561 in 3,914 posts in the 12 before (Welch test, $p = 0.207$).
3. Free learning resources are scattered. The user cannot tell which ones still work.
Our backtest shows the risk of false confidence. Twelve months ahead, a skill's share of job posts changes little. Simple persistence beat every trend model we tested. Varity gives a tested range, a plain Rising or Falling label, and a dated source on every line. The user decides.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Varity is a decision aid. It forecasts skill demand, ranks roles and builds a learning plan.

• Statistical forecast. We model each skill's monthly share of job posts on a logit scale. A rolling-origin backtest over 24 test months chose the model: "mean of the last 3 months", with a 12-month error of 1.66 percentage points (trend models: 2.13 and 2.27). A conformal band covered 93.8% of real values against an 80% target.
• Welch trend test. Labels each skill Rising, Falling or No clear change. The direction held in 67.2% of 720 backtest cases.

• Learned role profiles. Skill weights for 9 tech roles, learned from Hacker News posts that name exactly one role.
• A model that lost. A gradient-boosting model scored 2.06, worse than 1.66. We did not ship it.
• CV intake with Claude Haiku 5.5. The model only points at skills and gives a quote for each. Code drops any skill whose quote is not in the CV, word for word.

Only the CV intake and two optional steps use a language model. The route advisor re-orders roles by at most 2 places; the pathway coach re-orders steps within a stage and picks among listed resources. Code sets every number and date, checks every model answer and takes over when the model is off. "Try an example" needs no model.

Workshop skills used: GenAI APIs (Claude CV intake), database (Replit PostgreSQL for accounts and profiles), cloud hosting (Replit Autoscale deployment).

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

• Language and app: Python 3.12. FastAPI + uvicorn serve a static HTML/CSS/JavaScript web app (Flow map, CV & routes, Today, Pathway, Signals, Job forecast, Skills radar, Method).

• Data and statistics: pandas, NumPy, SciPy (Welch test), statsmodels (tested only); scikit-learn for the experiment (not shipped).
• Data sources: Hacker News "Who is hiring?" threads via the free Algolia HN Search API (94 threads, 48,508 posts, Jan 2019 – Oct 2026); Anthropic Economic Index (AI exposure); GPTs-are-GPTs, AIOE, Microsoft Working with AI (cross-check).
• Database: Replit PostgreSQL (accounts, saved profiles); SQLite fallback.
• Generative AI: Claude Haiku 5.5 (Anthropic API) for CV intake, route advisor and pathway coach, with a $1/day cap and a no-LLM fallback.
• Hosting: Replit Autoscale deployment, live at https://verity--eezhan0805.replit.app. Sign-in: Replit Auth (OIDC).

Block diagram:
Hacker News (Algolia API) → Skill tagger (40 skills) → Monthly demand → Forecast + band + Welch label → data files
Anthropic Economic Index → AI exposure + scenarios → data files
User CV → Claude Haiku 5.5 (quote-checked by code) → Role match → Pathway + daily plan
Data files + Role match + Pathway → FastAPI on Replit → Web app ↔ Replit Auth / PostgreSQL

Sponsor resources: Yes, Replit (hosting, Replit Auth sign-in, PostgreSQL, Secrets). We do not use Replit AI Integrations; Claude is called with our own Anthropic key. We do no GPU work, so we do not use RunPod.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise. Our user is Aina, a junior developer built from the Domain 1 brief, plus two more personas (Ben, Chloe).

Define. She needs a dated trend, a role match with a skill gap, and an ordered plan with free resources.
Ideate. LinkedIn and X break terms or hold personal data; JobStreet blocked plain requests with a bot challenge. We chose Hacker News hiring threads. We tried a machine-learning forecaster and dropped it when it lost.
Prototype. Varity, an eight-screen web app with a FastAPI backend.
Test. Over 700 automated tests and a backtest with no look-ahead. In Phase 2 we test with the three personas and volunteer students.
Deployment. Live on a Replit Autoscale deployment since 24 Oct 2026. If the server is unreachable, the page shows a dated offline snapshot and says so.
Hardest risk. Hacker News is US-heavy. The app says so; Malaysian sources are future work.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4.4: "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Supporting: SDG 8.2 (technological upgrading) and 8.5 (productive employment).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In Malaysia, only 26% of professionals have received formal training in AI tools, while 56% already use AI tools at work (Hays Asia Salary Guide 2025). MOSTI: more than 600,000 workers need reskilling within 3–5 years. Sources: Hays Asia Salary Guide 2025, as reported in People Matters, 'AI in Malaysian workplaces: a strategic outlook', https://sea.peoplemattersglobal.com/article/strategic-hr/ai-in-malaysian-workplaces-a-strategic-outlook-46436; MOSTI (Minister Chang Lih Kang), Malay Mail, 7 Jan 2025, https://www.malaymail.com/news/malaysia/2025/01/07/mosti-ai-disruption-looms-in-malaysia-600000-workers-must-reskill-in-three-to-five-years/162368; UN SDG targets: https://sdgs.un.org/goals
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. % of test users who finish at least one plan step using a link-checked free resource within 14 days (SDG 4.4). 2. % of test users who pick a target role after seeing the match score and skill gap (SDG 8.5). 3. % of test users who add at least one Rising or AI-complement skill to their plan (SDG 8.2).

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Targets (goals, not results): KPI 1 ≥ 50%; KPI 2 ≥ 80%; KPI 3 ≥ 60% of Phase 2 test users. Environmental note: CPU only, no model training in production.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☐ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☐ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it: None. The repository was created on 21 Oct 2026, after the build clock started (19 Oct 2026). All code, data and documents were written during the event.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI, uvicorn, Jinja2, pydantic, httpx, APScheduler	code	MIT / BSD-3-Clause
pandas, NumPy, SciPy, statsmodels	code	BSD-3-Clause
requests	code	Apache-2.0
pypdf, feedparser, Markdown, PyYAML	code	BSD / MIT
anthropic (Python SDK)	code	MIT
replit-object-storage	code	ISC

scikit-learn (experiment only, not shipped)	code	BSD-3-Clause
pytest (tests)	code	MIT
Hacker News 'Who is hiring?' threads (Algolia HN API)	dataset	No licence stated; only aggregate counts and thread links committed
Anthropic Economic Index (2026-06-26)	dataset	CC BY (data), MIT (code)
O*NET 31.0 Task Statements, O*NET-SOC crosswalk	dataset	CC BY 4.0
GPTs are GPTs ratings (OpenAI)	dataset	MIT
AIOE (Felten, Raj, Seamans)	dataset	No licence file; cited, not redistributed
Microsoft Working with AI scores	dataset	CC BY 4.0
Claude Haiku 5.5 (claude-haiku-5-5), Anthropic API	model	Anthropic commercial API terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns: Not applicable. Neither member has such an agreement.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Signature: Tay Ee Zhan (Team Leader) Goh Sheng Kuan (Member)

Date: 25 October 2026